

[21] Revision History

Version 1.0.0 - 2023-12-12

- Added
- TVS protection and termination switch to FD-CAN.
 - Low-side switched 12V 600 mA source for external fan.
 - LDO for analog supply.
 - One TVS diode per power connector.
 - Second on-board I2C magnetic encoder for disambiguation.
 - ESD protection to all interfaces.
 - Over-voltage protection on thermistor ADC inputs.
 - Pi filters to inputs of buck regulators and MCU analog supply.
 - Decoupling caps next to power pins of connectors.
- Changed
- CPH-CPL capacitor to 47 nF (gate driver).
 - FD-CAN transceiver IC.
 - FETs for top cooled variant.
 - Input power TVS diode to bidirectional.
 - Moved SOx low-pass filter to MCU section.
 - PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing.
 - RS422 pinout on connector.
 - Buck regulators to optimize for low noise.

Version 1.0.1 - 2024-01-25

- Added
- Controller target specifications.
 - Credits to moteus on cover page.
 - Optional RC-Snubber to power stage.
- Fixed
- Chassis guard ring to go around the board.
 - CAN and power TVS diodes now go to chassis.
 - Clearance between nets to respect IEC60664-1 where possible.
 - Comment on precharge.
 - Power TVS diode reference designator from "U" to "D".
- Changed
- 5V 300 mA buck converter with 600 mA version.
 - Chassis-GND capacitor by 1nF 1kV.

Version 1.0.2 - 2024-03-12

- Changed
- Power sequencing graph according to experimental data.

Version 1.1.0 - 2024-04-13

- Added
- RC snubber passive values.
- Fixed
- More vias for VBUS and LMR36006 GND pads.
 - Board version voltage reference from +3V3 to +A3V3.
- Changed
- Motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.

Version 1.1.1 - 2025-01-18

- Fixed
- Replace non-stocked components.

	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
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